Group code: TUE-16 **Student names:** Arijit Paul, Rishith Gupta, Arshit Singh, Prithvi Sangam Chakravarthy, Veeresh S K **EDL 2026**

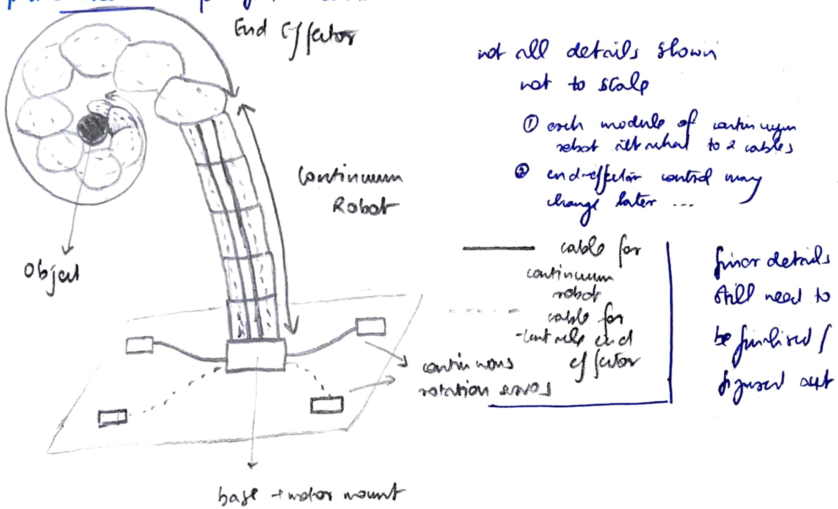
Project title: S-08 Continuum Robot with Tentacle End Effector **Date:** 20/01/2026

Use your notebook for discussions and rough work. Fill out this sheet after working individually and discussing within your team.

1. **In simple words, describe what you are going to build in your project, what its purpose is, and how it will function.** Be as detailed as possible, covering all the major aspects of your project.
    a. What is the main goal of your project?
    b. What problem does it solve, and how?
    c. Who will use your project, and in what context?

Draw a pencil sketch of what your project will look like at the end of the course, for final demo.

pencil sketch for final demo



a. main goal: apply the prior knowledge, skills, ~~along with~~ dealing with mechanical components to make a tentacle end effector with continuum robot
[taking best of 2 worlds 'rigid' robotics and 'soft' robotics where rigid ones fail and soft robots lack control

b. soft robotics: helps grip irregular objects with multiple degrees of freedom
adaptive gripping of objects, reach cluttered spaces, ~~and~~ lightweight (so easy to handle)
"Rigidity trap"

c. multipurpose robots for industrial automation, health-care assistance, search-rescue operations and deep-space / underwater exploration.
many possible applications

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2. **Draw a block diagram of your project.** Create a visual representation showing the key components or subsystems of your project. For each block in the diagram, briefly explain its main function and how it fits into the overall system.
   a. What are the main subsystems or modules of your project?
   b. How do they interact with each other?

Tentacle End Effector
+
Continuum Robot
((main entity, final product))

ELECTRICAL

→ optimum servo (continuous rotation) selection and connections within structure

→ power management and supply to various components like servos, MCU, servo driver etc.

→ coding the MCU to obtain desired functionality from the system

→ bread boarding + PCB design for the entire circuitry

SOFTWARE

→ interfacing the microcontroller to operate the actuators, properly generating PWM signals

→ laptop-based control of the final product for demo

→ GUI for the same which is both interactive + intuitive

MECHANICAL

→ link-based tentacle structure with spiral geometry

→ continuum arm with independent control for controlling different links

→ cables routing through the structure to the actuator for proper movement

→ base to support the whole structure and mounts for the actuators (servos)

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4. **What are the unknowns or uncertainties in this project?** Identify aspects of your project that you are uncertain about or that require further research. This may include areas where you know what you need to do but are unsure how to approach it. **Use more sheet(s) if needed.**
   a. What technical challenges or questions are you facing?
   b. Are there any assumptions you must make in order to move forward?

a) ① mechanical components fit ? ② proper movement without friction ? ③ enough torque delivered by continuous 360 rotation servos
④ hands-off control from laptop ? ⑤ actual mechanism for the end-effector (going with leg. spiral for now)

few questions / challenges we have now ↑ , might add more as we progress, remove as resolve them

b) assumptions (i) all electrical components will function properly (ii) some mathematical while deriving stuff for mech. design
(iii) 3-D printing would be correct, and structure won't be heavy (iv) ← for sets determining required torque for actuator

Other things to consider from now until Milestone 1 deadline:

5. **Roles and Responsibilities:** How will the work be divided among team members? Assign specific tasks and responsibilities to each team member. Be clear about who is responsible for each part of the project.
   a. Who will work on which blocks or subsystems? In what capacity (hardware/software; design/implementation/testing etc.)
   b. What are the deadlines for each task?
   c. How will the team communicate and coordinate to ensure everyone is on track?

6. **Next Steps: What is your plan for the next phase of the project?** Outline what needs to be done in the short-term to move forward.
   a. What are the immediate next tasks or priorities?
   b. Are there any dependencies between tasks? How will you handle these interdependencies?
   c. What resources or materials do you need to proceed?

7. **Feedback and Collaboration: How will you gather feedback and collaborate during the project?** Describe how your team plans to share progress, give and receive feedback, and collaborate throughout the course of the project.
   a. How often will you check in with your team members?
   b. Will you conduct regular brainstorming or review sessions?

5\. a. Right now, work is being done on Mechanical + Electrical subsystem only, software will come up during integration!

Mechanical → Tentacle End-Effector — Rishith + Arshit
Mechanical → Continuum — Veeresh

Electrical — Arijit + Prithvi

b. Deadlines for each task designated as per Gantt chart, refer that ~~please~~ for more details

c. Team will communicate and coordinate over Whatsapp and One Note to ensure everyone is on-track. Veeresh will manage and ensure all tasks are done overall!

6\. a. immediate next tasks + priorities
→ finalize design for tentacle end effector
→ prototyping for continuum robot
→ electronic circuit design + connections for servos

b. Yes, there are obviously dependencies among the subsystems. To handle these team members will coordinate and consult with each other regularly ensuring the other guy is updated with the latest information

c. ① materials mentioned in BOM
② guidance of WEL RAs, staff + Faculty Advisor

will add if anything else arises!

7.

a. we plan to have upto 2-3 weekly meets for the whole team other than distributed subsystem-specific meets which will happen throughout the week

b. yes, we will and we do so, all decisions and choices are made through such regular brainstorming or review sessions